

ROUTER CONFIGURATION GUIDE

Project: University Campus Networking System

Platform: Cisco Packet Tracer 8.2.2

Routing Protocol: RIP Version 2

1. Cloud Router Configuration

Purpose

The Cloud Router provides connectivity between the simulated external network and the university campus. It also exchanges routing information with the Main Campus Router using RIP Version 2.

Configuration Summary

Enabled Interfaces

- GigabitEthernet0/0
- Serial0/1/0

Assigned IP Addresses

Interface	IP Address	Purpose
GigabitEthernet0/0	20.0.0.1/30	External Cloud Network
Serial0/1/0	10.10.10.6/30	Connection to Main Campus Router

Routing Protocol

- RIP Version 2
- Advertised Networks:
 - 20.0.0.0
 - 10.0.0.0

Verification

The router interfaces were enabled successfully, and routing updates were exchanged with the Main Campus Router.

2. Main Campus Router Configuration

Purpose

The Main Campus Router acts as the central routing device of the university network. It performs:

- Inter-VLAN Routing (Router-on-a-Stick)
- DHCP Relay
- Dynamic Routing
- Communication with the Branch Campus
- Communication with the Cloud Router

VLAN Gateway Configuration

VLAN	Department	Gateway
10	Administration	192.168.1.1
20	Human Resources	192.168.2.1
30	Finance	192.168.3.1
40	Business	192.168.4.1
50	Electronics & Communication	192.168.5.1
60	Academic Department	192.168.6.1
70	Student Department	192.168.7.1
80	IT Department	192.168.8.1

WAN Interfaces

Interface	IP Address	Connected To
Serial0/1/0	10.10.10.5/30	Cloud Router
Serial0/1/1	10.10.10.1/30	Branch Campus Router

Services

- Router-on-a-Stick
 - DHCP Relay using IP Helper Address
 - RIP Version 2
 - VLAN Gateway Routing
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Routing Networks

The router advertises:

- 10.0.0.0
- 192.168.1.0
- 192.168.2.0
- 192.168.3.0
- 192.168.4.0
- 192.168.5.0
- 192.168.6.0
- 192.168.7.0
- 192.168.8.0

Verification

Successful routing was verified using:

- show ip route
- show ip interface brief
- End-to-end ping tests between Main Campus and Branch Campus

3. Branch Campus Router Configuration

Purpose

The Branch Campus Router connects the remote campus with the Main Campus through a serial WAN link. It provides gateway functionality for the Branch Campus VLANs and exchanges routes using RIP Version 2.

VLAN Gateway Configuration

VLAN	Department	Gateway
90	Staff	192.168.9.1
100	Student Laboratory	192.168.10.1

WAN Interface

Interface	IP Address	Connected To
Serial0/1/0	10.10.10.2/30	Main Campus Router

Routing Protocol

- RIP Version 2

Advertised Networks

- 10.0.0.0
 - 192.168.9.0
 - 192.168.10.0
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Services

- Inter-VLAN Routing
 - DHCP Gateway
 - Dynamic Routing
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Verification

Successful connectivity was confirmed through:

- show ip route
 - show ip interface brief
 - Branch-to-Main Campus ping tests
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4. Configuration Validation

The following verification commands were used to validate the network configuration:

show running-config

show ip interface brief

show ip route

These commands were executed on all routers to verify:

- Interface status
 - IP addressing
 - Routing table
 - RIP updates
 - Connected and learned networks
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5. Overall Outcome

The router configuration successfully established communication between:

- Cloud Network
- Main Campus
- Branch Campus

The implemented configuration supports:

- Dynamic routing using RIP Version 2
- Router-on-a-Stick Inter-VLAN Routing
- DHCP Relay
- Internal Server Connectivity
- End-to-End Network Communication
- DNS, HTTP, FTP and Email Services.